

Task Description

Goal: place the red cube on top of the green cube and then open the gripper.

API Documentation

```
def solve_ik(position, quaternion_wxyz):  
    ...  
  
def move_to_joints(joints):  
    ...  
  
def sample_grasp(depth, intrinsics, seg_mask):  
    ...  
  
def point_prompt_molmo(rgb, text_prompt):  
    ...  
  
def segment_sam3_text_prompt(rgb, text_prompt):  
    ...  
...
```

Skill Library

```
def rot_matrix_to_quat(rotation_matrix):  
    ...  
  
def depth_to_pointcloud(depth, intrinsics):  
    ...  
  
def transform_pts(points, transform_matrix):  
    ...  
  
def select_top_down_grasp(grasps, scores, ...):  
    ...  
...
```

Parallel Query

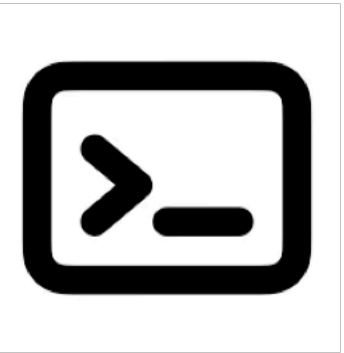
 Coding Agent

 Coding Agent

 Coding Agent

Candidate
Code
Solutions

 Ensemble Agent



CaP-Agent0

Env.
Feedback

Env.
Description

 Visual Differencing Model (VDM)

In the center of the scene, there are two stacked cubes: a green cube rests directly on top of a red cube. The robot's gripper is partially visible at the top edge of the frame, currently in an open state.

Stdout:
Stage 1: Unstacking Green Cube
...
Stderr:
...

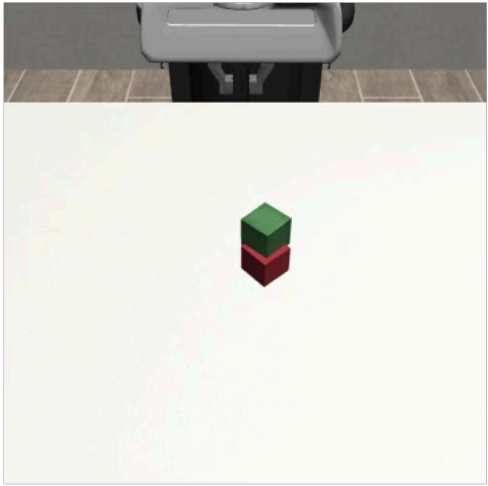
Runtime Outputs

Generated Python Code

```
import numpy as np  
  
def get_best_mask(masks):  
    """Selects the best mask based on score and area."""  
    ...  
  
def main():  
    # --- STAGE 1: UNSTACK GREEN CUBE ---  
    print("Stage 1: Unstacking Green Cube")  
    obs = get_observation()  
    cam = obs["robot0_robotview"]  
    rgb = cam["images"]["rgb"]  
    depth = cam["images"]["depth"]  
    K = cam["intrinsics"]  
    T_world = cam["pose_mat"]  
  
    # Segment Green Cube (Top)  
    green_res = get_best_mask(  
        segment_sam3_text_prompt(rgb, "green cube")  
    )  
    ...
```

Python Sandbox

Robot Environment



robosuite

Coding Primitives

 Molmo

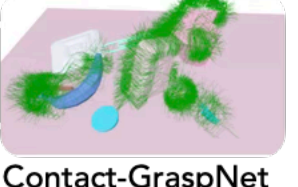
 OpenCV

 NumPy

 PyRoki

 SciPy

 OPEN3D

 Contact-GraspNet

 SAM 3